

Seokha Moon

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About Me

I completed my Ph.D. degree at Korea University's Vision & AI Lab, advised by Prof. Jungbeom Lee and Prof. Jinkyu Kim, and currently work as a AI planning researcher in KakaoMobility's Autonomous Driving Development Team. My research focuses on **autonomous driving and robotics across perception, prediction, and planning**. I have led first-author projects on multi-camera 3D object detection, streaming 3D occupancy prediction, VLM-guided trajectory prediction, and causality-aware end-to-end planning.

My current interests include reliable closed-loop planning and Vision-Language-Action (VLA) models that connect perception, reasoning, and decision-making.

Education

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| Ph.D. | Korea University , Artificial Intelligence Applications | Sep. 2022 to Aug. 2026 |
| | <ul style="list-style-type: none"> GPA: 4.36/4.5 Advisors: Prof. Jungbeom Lee and Prof. Jinkyu Kim | |
| BS | Yonsei University , Computer Science | Mar. 2016 to Aug. 2022 |
| | <ul style="list-style-type: none"> GPA: Overall 3.45/4.3 Major 3.97/4.5 Coursework: Operating System, Computer Architecture, Algorithms, Data Structures, Computer Graphics, Object-Oriented Programming, Artificial Intelligence, Computational Theory. | |

Experience

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| KakaoMobility , Autonomous Driving Development Team, Planner Developer | Gyeonggi-do, South Korea
Apr. 2026 to Present |
| <ul style="list-style-type: none"> Developing planning algorithms that transform perception, prediction, map, and vehicle-state inputs into safe and natural autonomous driving behavior and trajectories. Analyzing scenario-level failures and improving behavior decision, trajectory generation, real-time integration, and verification logic for real-vehicle deployment. | |
| NAVER LABS , Robot & AD Localization, Research Intern | Gyeonggi-do, South Korea
Sep. 2024 to Mar. 2025
6 months |
| <ul style="list-style-type: none"> Led a streaming 3D occupancy prediction project for camera-based 3D scene understanding, defining the latency, memory, and coordinate-alignment issues in multi-frame fusion. Designed stream-based voxel aggregation and query-guided refinement to preserve dense spatial representations while improving real-time efficiency and dynamic-object modeling. | |
| 42dot , Motion Prediction Team, Research Intern | Seoul, South Korea
Aug. 2022 to Dec. 2022
4 months |
| <ul style="list-style-type: none"> Developed interaction-aware trajectory prediction models for multi-agent intersection scenarios. Designed a future-aware teacher-student distillation framework that transfers interaction knowledge from privileged future observations to an efficient inference-time model. | |
| Korea Univ. Vision & AI Lab , Research Intern | Seoul, South Korea
Jan. 2022 to Aug. 2022
9 months |
| <ul style="list-style-type: none"> Worked on multi-camera 3D object detection and occupancy prediction, including ORA3D and Waymo occupancy-challenge baselines. Built perception data processing and evaluation pipelines for 3D detection and occupancy prediction experiments. | |

- Implemented an attention-based few-shot face detection algorithm for limited-data settings.
- Built and curated a face dataset and designed augmentation pipelines covering scale, crop, color, and blur variations.

Publications

- RECAST: Recasting Vision-Language Semantics into an Actionable Cost Map for Robot Navigation** 2026
Incheol Cho*, Jintae Park*, Jinkyu Kim, Jungbeom Lee, Jaegul Choo[†], Seokha Moon[†]
We present RECAST, a robot navigation framework that combines the reasoning of a vision-language model with the spatial grounding of vision foundation models to build an actionable cost map, judging which surfaces are traversable, which objects pose a risk, which heading to prefer, and which gaps are passable.
Under Review
- Causality-Aware End-to-End Autonomous Driving via Ego-Centric Joint Scene Modeling** 2026
Seokha Moon, Minseung Lee, Joon Seo, Jinkyu Kim, Jungbeom Lee
We propose CaAD, an end-to-end autonomous driving framework that learns ego-centric joint-causal dependencies between the ego vehicle and interaction-relevant agents, then aligns the policy using planning-oriented closed-loop feedback.
arXiv Preprint
- ALIGN: Advanced Query Initialization with LiDAR-Image Guidance for Occlusion-Robust 3D Object Detection** 2026
Janghyun Baek, Mincheol Chang, Seokha Moon, Seung Joon Lee, Jinkyu Kim
We introduce an object-aware, LiDAR-image guided query initialization strategy that significantly improves detection of occluded and crowded objects without modifying detector architectures.
arXiv (Under Review)
- Image-Guided Semantic Pseudo-LiDAR Point Generation for 3D Object Detection** 2026
Minseung Lee, Seokha Moon, Seung Joon Lee, Reza Mahjourian, Jinkyu Kim
We introduce ImagePG, an image-guided pseudo-LiDAR point generation method that improves 3D object detection for sparse and distant objects while reducing false positives.
Winter Conference on Applications of Computer Vision (WACV)
- SUPER-AD: Semantic Uncertainty-aware Planning for End-to-End Robust Autonomous Driving** 2025
Wonjeong Ryu, Seungjun Yu, Seokha Moon, Hojun Choi, Junsung Park, Jinkyu Kim, Hyunjung Shim
SUPER-AD introduces uncertainty-aware BEV planning with lane-following priors for safer and more robust camera-only end-to-end autonomous driving.
arXiv (Under Review)
- Streaming Dense Voxel Representations for 3D Occupancy Prediction** 2026
Seokha Moon, Janghyun Baek, Yujin Jeong, Daewon Chae, Giseop Kim, Jungbeom Lee, Jinkyu Kim, Sunwook Choi
We propose StreamOcc, a dense voxel streaming framework for accurate and efficient 3D occupancy prediction, reducing warping artifacts while selectively injecting instance-level query features into occupied voxel regions.
Work done during an internship at NAVER LABS.
European Conference on Computer Vision (ECCV)
- Who Should Have Been Focused: Transferring Attention-based Knowledge from Future Observations for Trajectory Prediction** 2024
Seokha Moon, Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim
We propose an attention-based teacher-student method that distills future-aware interaction knowledge to improve multi-agent trajectory prediction, yielding safer and more human-like predictions.
International Conference on Pattern Recognition (ICPR)

VisionTrap: Vision-augmented Trajectory Prediction Guided by Textual Descriptions <u>Seokha Moon</u> , Hyun Woo, Hongbeen Park, Haeji Jung, Reza Mahjourian, Hyung-gun Chi, Hyerin Lim, Sangpil Kim and Jinkyu Kim We introduce a novel approach incorporating visual features from surround-view cameras and textual descriptions for trajectory prediction. European Conference on Computer Vision (ECCV)	2024
Learning Temporal Cues by Predicting Objects Move for Multi-camera 3D Object Detection <u>Seokha Moon</u> , Hongbeen Park, Jungphil Kwon, Jaekoo Lee, and Jinkyu Kim We propose a Detection After Prediction (DAP) method to explicitly learn the temporal cues. IEEE International Conference on Robotics and Automation (ICRA)	2024
BEVMap: Map-Aware BEV Modeling for 3D Perception Mincheol Chang, <u>Seokha Moon</u> , Reza Mahjourian and Jinkyu Kim IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)	2024
Enhancing Trajectory Prediction Accuracy with Goal Location and Lane Information <u>Seokha Moon</u> and Jinkyu Kim We introduce a method to predict the goal location and interact with lane to navigate towards the predicted goal. Conference on The Institute of Electronics and Information Engineers (IEIE) Oral	2023
RUFI: Reducing Uncertainty in behavior prediction with Future Information <u>Seokha Moon</u> , Sejeong Lee, Hyun Woo, Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Vision-Centric Autonomous Driving	2023
ORA3D: Overlap Region Aware Multi-view 3D Object Detection Wonseok Roh, Gysam Chang, <u>Seokha Moon</u> , Giljoo Nam, Chanyoung Kim, Younghyun Kim, Jinkyu Kim and Sangpil Kim British Machine Vision Conference (BMVC)	2022

Teaching

Mathematics for Computer Science (COSE111)	Spring 2024
Algorithm (COSE214)	Fall 2023
Algorithm (COSE214)	Fall 2022

Honors And Awards

First Prize of Computer Science , Yonsei University	Aug. 2021
HUAWEI scholarship , Talent Development Foundation	Nov. 2020

Skills

Languages: Python, C++, C, JavaScript

FRAMEWORKS & TOOLS: PyTorch, OpenCV, TensorFlow, Git, Linux, CUDA, $\text{\LaTeX}$